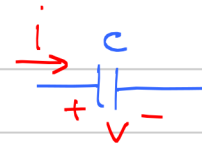




Capacitor in phasor domain

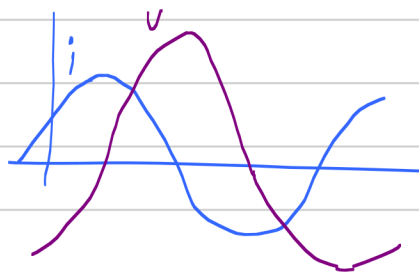
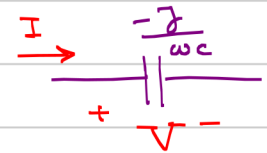
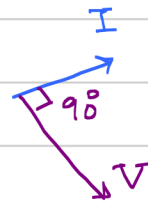


$$v = V_m \cos(\omega t + \theta_v) \Rightarrow V = V_m \angle \theta_v$$

$$i = c \frac{dv}{dt} = c V_m [-\omega \sin(\omega t + \theta_v)] = \omega c V_m \cos(\omega t + \theta_v + 90^\circ)$$

$$\Rightarrow I = \omega c V_m e^{j\theta_v} e^{j90^\circ} = j\omega c V_m e^{j\theta_v} = j\omega c V$$

$$Z_c = \frac{V}{I} = \frac{1}{j\omega c} = \frac{-j}{\omega c}$$

 $\Rightarrow$ 

capacitor current leads the voltage by 90° .

In phasor domain all circuit elements behave as resistors, having a linear relationship between their voltages and currents.

The ratio of V/I is called impedance.

$$Z = \frac{V}{I} = R + jX$$

$\uparrow$ impedance (Ω) $\uparrow$ resistance (Ω) $\uparrow$ reactance (Ω)

circuit element	Impedance (Z)	Reactance (X)	$\theta_v - \theta_i$
resistor	R	—	0
inductor	$j\omega L$	ωL	90°
capacitor	$\frac{-j}{\omega c}$	$\frac{-1}{\omega c}$	-90°

To analyze ac circuits, first transform all circuit elements into phasor domain. Then treat the circuit as a resistive circuit.

Ex. The voltage across a 5MF capacitor is $V = 30 \cos(4000t + 25)$.

Find a) capacitive reactance b) impedance c) phasor current I

d) time domain current $i(t)$.

$$a) X_c = \frac{-1}{\omega c} = \frac{-1}{(4000)(5 \times 10^{-6})} = -50 \Omega$$

$$b) Z_c = \frac{-j}{\omega c} = -j50 \Omega$$

$$c) I = \frac{V}{Z} = \frac{30 \angle 25}{-j50} = \frac{30 \angle 25}{50 \angle -90} = 0.6 \angle 115^\circ A$$

$$d) i(t) = 0.6 \cos(4000t + 115) A$$

Are KVL and KCL valid in phasor domain? We need to prove that sum of the two

sinusoidal functions has the same result as the sum in phasor form.

time domain:

$$V = V_1 + V_2 = V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2)$$

$$= V_{m1} \cos \omega t \cos \theta_1 - V_{m1} \sin \omega t \sin \theta_1 + V_{m2} \cos \omega t \cos \theta_2 - V_{m2} \sin \omega t \sin \theta_2$$

$$= \underbrace{(V_{m1} \cos \theta_1 + V_{m2} \cos \theta_2)}_A \cos \omega t - \underbrace{(V_{m1} \sin \theta_1 + V_{m2} \sin \theta_2)}_B \sin \omega t$$

$$= A \cos \omega t - B \sin \omega t = \sqrt{A^2 + B^2} \left[\underbrace{\frac{A}{\sqrt{A^2 + B^2}}}_{\cos \phi} \cos \omega t - \underbrace{\frac{B}{\sqrt{A^2 + B^2}}}_{\sin \phi} \sin \omega t \right]$$
$$= \sqrt{A^2 + B^2} \cos(\omega t + \phi) \quad , \quad \phi = \tan^{-1} \frac{B}{A} \quad (1)$$

phasor domain :

$$V = V_1 + V_2 = V_{m1} \angle \theta_1 + V_{m2} \angle \theta_2 = V_{m1} \cos \theta_1 + j V_{m1} \sin \theta_1 + V_{m2} \cos \theta_2 + j V_{m2} \sin \theta_2$$
$$= \underbrace{(V_{m1} \cos \theta_1 + V_{m2} \cos \theta_2)}_A + j \underbrace{(V_{m1} \sin \theta_1 + V_{m2} \sin \theta_2)}_B$$

$$V = A + jB = \sqrt{A^2 + B^2} \angle \underbrace{\tan^{-1} \frac{B}{A}}_{\phi} = \sqrt{A^2 + B^2} \angle \phi$$

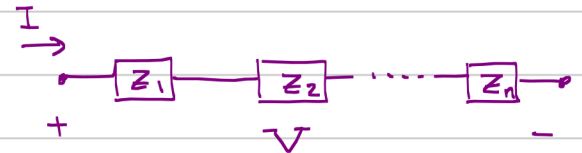
inverse phasor transform: $V = \sqrt{A^2 + B^2} \cos(\omega t + \phi)$ ②

① = ② $\Rightarrow$ Kirchhoff's laws are valid in phasor form.

Series and parallel combination of impedances

Series combination:

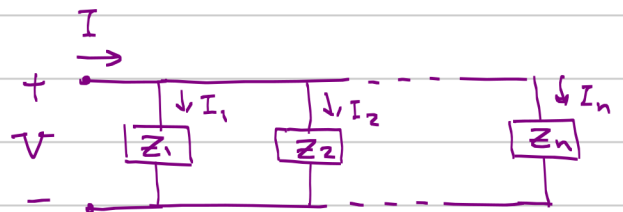
$$V = Z_1 I + Z_2 I + \dots + Z_n I$$



$$= (Z_1 + Z_2 + \dots + Z_n) I \Rightarrow Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \dots + Z_n$$

Parallel combination:

$$I = I_1 + I_2 + \dots + I_n$$



$$I = \frac{V}{Z_1} + \frac{V}{Z_2} + \dots + \frac{V}{Z_n}$$

$$I = \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n} \right) V = \frac{1}{Z_{eq}} V$$

$$\Rightarrow Z_{eq} = \frac{V}{I} = \frac{1}{\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}}$$

Ex. Find $v(t)$ and $i(t)$.

$$V_s = 5 \sin 10t = 5 \cos(10t - 90^\circ)$$

$$\Rightarrow V_s = 5 \angle -90$$

$$Z_L = j\omega L = j(10)(0.2) = j2 \Omega$$

$$Z_{eq} = R + Z_L = 4 + j2 \Omega$$

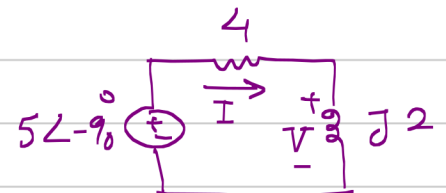
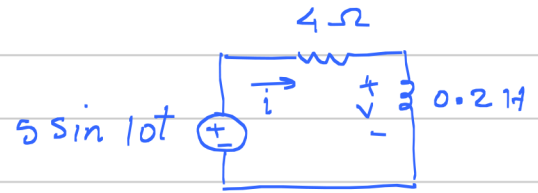
$$I = \frac{V_s}{Z_{eq}} = \frac{5 \angle -90}{4 + j2} = \frac{5 \angle -90}{\sqrt{4^2 + 2^2} \angle \tan^{-1} \frac{2}{4}} = \frac{5 \angle -90}{4.47 \angle 26.56}$$

$$I = 1.118 \angle -116.56 \Rightarrow i(t) = 1.118 \cos(10t - 116.56^\circ)$$

$$V = Z_L I = (j2)(1.118 \angle -116.56) = (2 \angle 90)(1.118 \angle -116.56)$$

$$= 2.236 \angle -26.56$$

$$\Rightarrow v(t) = 2.236 \cos(10t - 26.56)$$



phasor domain

Ex. Find $v_1(t)$ and $v_2(t)$ using nodal analysis.

$$Z_L = j\omega L = j(2)(2) = j4 \Omega$$

$$Z_C = \frac{-j}{\omega C} = \frac{-j}{2 \times 0.2} = -j2.5 \Omega$$

node 1:

$$-10 \angle -90^\circ + \frac{V_1}{2} + \frac{V_1 - V_2}{-j2.5} = 0$$

$$(5 + j4) V_1 - j4 V_2 = 100 \angle -90^\circ \quad (1)$$

node 2:

$$\frac{V_2 - V_1}{-j2.5} + \frac{V_2}{j4} + \frac{V_2 - 3V_1}{4} = 0$$

$$-(7.5 + j4) V_1 + (2.5 + j1.5) V_2 = 0 \quad (2)$$

$$V_1 = \frac{2.5 + j1.5}{7.5 + j4} V_2 = 0.343 \angle 2.89^\circ V_2$$

①:

$$(5 + j4) (0.343 \angle 2.89^\circ) V_2 - j4 V_2 = 100 \angle -90^\circ$$

$$V_2 = 33 \angle -32.9^\circ \Rightarrow v_2(t) = 33 \cos(2t - 32.9^\circ)$$

$$V_1 = 11.3 \angle -30^\circ \Rightarrow v_1(t) = 11.3 \cos(2t - 30^\circ)$$

